

اختبار نظم التشغيل - الترم الثاني ١٤٣٩ هـ / ١٤٤٠ هـ

Section 1 (3 Marks): OS Concepts and Definitions

Choose the correct answer. Questions 1-12 each has 0.25 Mark

1. The process long-term Scheduler is responsible for managing _____ effectively?
☒ a- CPU
☐ b- I/O devices
c- Mother Board
d- Non-of-the above
2. The process short-term scheduler _____
a- is called frequently
b- needs to decide very fast.
c- controls which process can enter the running state
☒ d- All f the above
3. The scheduler for a Batch System does not make use of the to switch the running process from running to ready state.
a- Input devices
☒ b- Clock
c- Memory
d- Non-of-the above
4. When a user process calls an operating system call, _____ occurs.
a- interrupt
☒ b- trap
c- exception
d- None of the above
5. memory technique is used to increase the degree of multiprogramming.
☒ a- Swapping
b. Scheduling
c. Dispatching
d. Context switch
6. After the completion of the DMA transfer, the processor is notified by
a- Acknowledge signal
☒ b- Interrupt signal
c- WMFC signal
d- None of the mentioned
7. Each hardware device connected to a computer system is assigned a unique number to identify it when interrupts occurs. The operating system uses this number to find out the address of the corresponding
a- Interrupt context switch
b- Interrupt handler routine
c- Interrupt vector
d- Dispatcher
8. A process enters the waiting queue when
a- it finishes its time quantum.
☒ b- it requests I/O operation.
c- interrupt occurs.

d- None of the above.

9. When a process uses blocking send system call in a message passing system, the kernel puts the calling process in state until a message is received

- a- new
- ☒ b- waiting ✓
- c- running
- ☒ d- ready X

10. Batch system can NOT be used for?

- a- Applications that run one after the other
- b- Applications that don't allow sharing of computer resources with others.
- ☒ c- Interactive application
- d- non-of-the-above

11. What is not true with Timesharing (multitasking) system?

- a- Operating system try to utilize the computer hardware resources efficiently.
- b- Operating system gives each process the CPU for certain period of time.
- ☒ c- Application couldn't know the time when they will get running.
- d- Application could not cooperate together in executing a common task.

12. In the producer-consumer problem with unbounded buffer

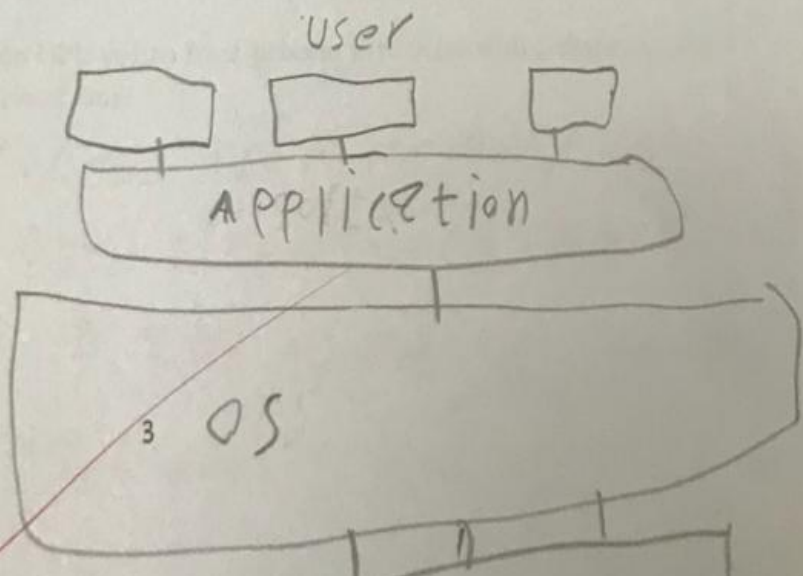
- ☒ a- The Producer can block but consumer not ✓
- b- The Consumer can block but producer not
- c- Both Producer and Consumer can never block
- ☒ d- Bother Producer and Consumer can Block X

Section 2 (12Marks):

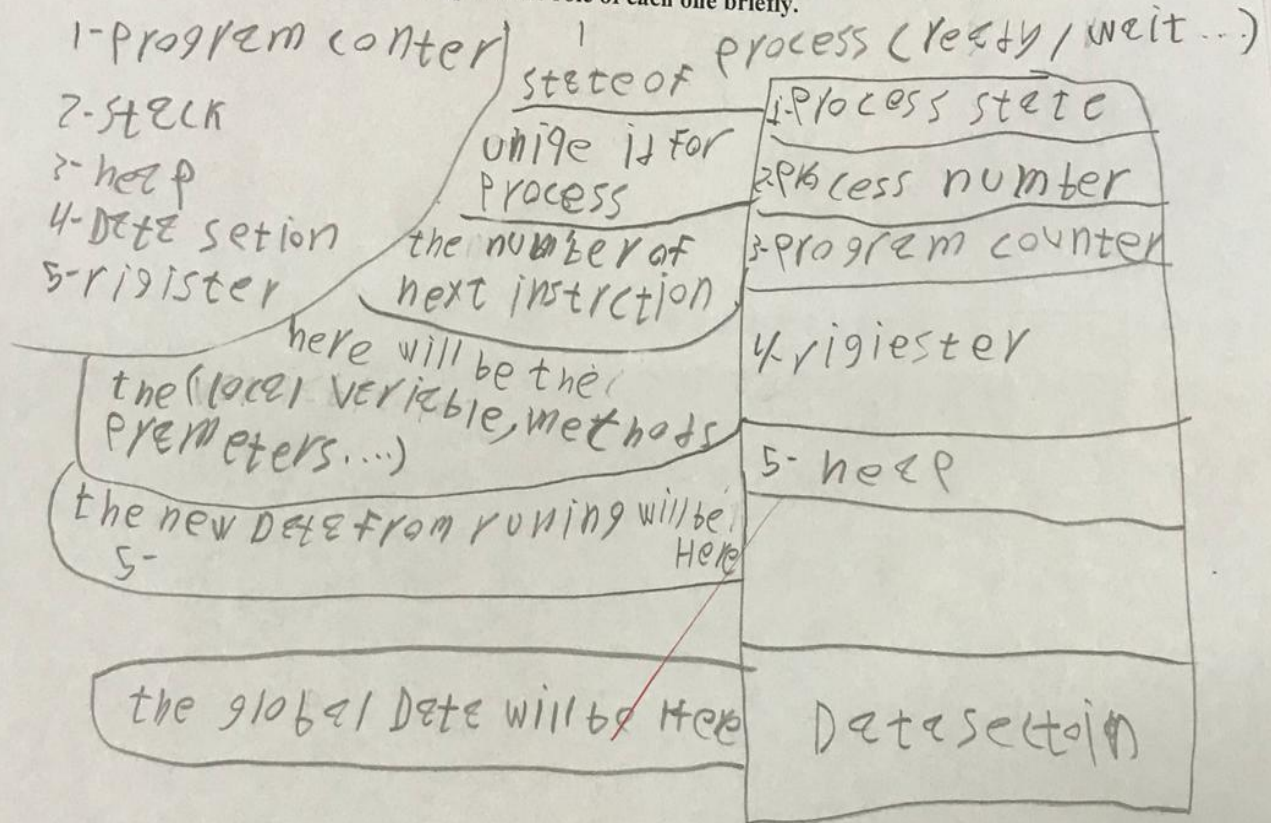
Question 13-20 has 1.5 Mark.

13. What is the role of the operating system? Explain components of computer system structure with a diagram.

operating system is resource allocator
which manage between hardware, software
and user, and it control system and it have
abstraction hide details to make more convient
interface



14. Mention 5 process attributes that are stored in the process control block (PCB) data structure of any process? Explain the role of each one briefly.



15. What is a context-switch? Explain CPU switch from process to process with a diagram. Does the size of the process affect the overhead time?

it's lost the process from memory
and of course the size affect on time of it
for management if the time is 100 m/s
and the manage take 20 m/s it will
be take it from its time

16. Explain how the operating system scheduler uses a timer to prevent a process to run long period than it is allowed or deserved?

Not allowed, because each process has same period and one not complete will set as ready and wait for his turn again

~~1/2~~

17. Explain the difference of the "kernel mode" and "user mode" during the execution of a process.

this is dual mode for protect CPU and process

kernel mode: when application need service by system call.

user mode: ~~the~~ when the system work for application⁵

~~1/2~~

18. Mention the role of the Process "text segment" and "data segment"

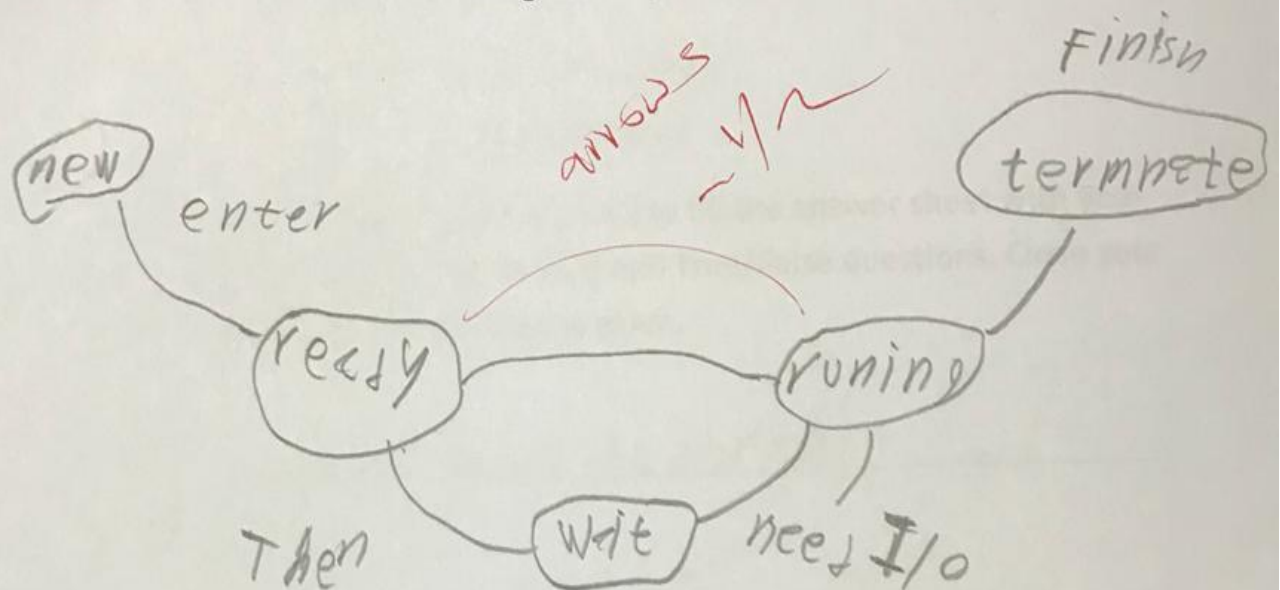
text segment: Provide the code
executed

data segment: Provide the data needed
for process

19. Explain the data transfer models of inter-process communication (IPC). Make sure to clarify the role of the operating system in each.

1- Pass message	2- shared memory
- Here need to copy data	- copy is optional
- involve kernel	- no need to involve kernel
- Easy to implement	- faster
Also slower for connected computer	Fast for single computer

20. Explain process state diagram. How many processes are in the running state? Single or multiple? What about ready and waiting states?



single is running (as some single processor)

ready and wait each one will be in queue